

ControlPlane.ai

Enterprise AI Governance & Safety Gateway — Technical README & System Architecture

1. Executive Overview & System Architecture

ControlPlane.ai is an inline enterprise AI governance and safety gateway designed to sit between AI applications and Large Language Model (LLM) providers/agents. Rather than relying on unmonitored model output or raw prompt completion, ControlPlane enforces organizational policies, scrutinizes request/response payload claims, gates high-risk tool calls, and tracks compliance audit logs with cryptographic hash chaining.

Flow Architecture:
Application Client → **ControlPlane Gateway** → **Upstream AI Model** → **Response Analysis** → **Policy Engine** → [ALLOW / MODIFY / ESCALATE / BLOCK] → **Application Client**

Component	Technical Specification & Role
Adaptive Scrutiny	Tier 0 cheap-gate heuristic pre-filter → Tier 1 full pipeline claim verification & PII detection.
Claim Verification	NLP Claim Extractor & Verification against knowledge corpus (SUPPORTED / CONTRADICTED / UNVERIFIABLE).
Taint Propagation	Multi-turn conversation state tracking; flags claims and prevents unverified data from triggering downstream tool calls.
Surgical Remediation	Per-claim remediation (Hedge, Redact, Remove, Cite, Escalate) instead of total response blockage.
Tool-Call Gating	Intercepts high-consequence tools (e.g. money movement, database updates) based on taint state and permission level.
Cost Circuit Breaker	Per-tenant rate and token circuit breaker to prevent denial-of-wallet and quota exhaustion spikes.
Audit Ledger	Cryptographically hash-chained immutable audit log for complete regulatory compliance and forensics.

2. Core Technical Capabilities & Implementation Status

All 5 project development phases are fully implemented, benchmarked, and test-verified:

- **OpenAI API Compatible Gateway:** Transparent drop-in replacement endpoint (`http://localhost:8000/v1``) with full request/response interception.
- **PII & Safety Scrutiny:** Multi-label risk classification (PII regex detection, claim hallucination checking, toxicity & prompt injection filtering).

- **Dynamic Risk Appetite Control:** Live tenant-level risk appetite slider allowing real-time adjustment of governance strictness without service restarts.
- **Human-in-the-Loop & Recalibration:** Interactive review interface with automated threshold-triggered recalibration recommendations.
- **Management Dashboard Console:** Real-time React + FastAPI management console connected to SQLite/PostgreSQL audit ledgers.

3. Benchmark & Verification Summary

ControlPlane has been rigorously validated through automated suite tests and a 400-item synthetic evaluation benchmark:

Evaluation Metric	Measured Result	Impact & Notes
Automated Test Suite	169 / 169 Passing	100% test coverage across unit & integration scenes.
Synthetic Traffic Replayer	10,000 requests / 100s	Demonstrated high-throughput audit ledger logging.
Calibration Error (ECE)	Expected Error < 0.05	Measured alignment between risk scores and true outputs.
Scrutiny Performance	Adaptive Routing	Balances latency (Tier 0 cheap gate) and accuracy (Tier 1).

4. Quick Start & Execution Guide

```
# 1. Environment Setup & Installation
python3 -m venv .venv && source .venv/bin/activate
pip install -e "[dev]"

# 2. Launch Gateway Service
uvicorn gateway.main:app --port 8000 --reload

# 3. Run Validation Test Suite & Benchmark
pytest
python -m bench.harness.run_benchmark

# 4. Launch Governance Console & Dashboard
uvicorn console.backend.main:app --port 8001
cd console/frontend && npm install && npm run dev
```

5. Official Repository Reference & Project Team

Public GitHub Repository: <https://github.com/MohithChandra07/Control-Plane-AIC>
Submission Track: Accenture Innovation Challenge 2026 — Prototype Development

Project Team & Roles:

- **Mohith Chandra:** Team Lead
- **Monal Gupta:** Core NLP & Fullstack Developer
- **Veda Vikas:** System Architect & Developer